

CHE 313

Exam 3

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1)

 $x_0 = 0.012 \text{ mol\% NB}$
 $L = 1 \text{ kmol/min}$

 $T_{sys} = 100^\circ\text{C}$
 $\frac{L}{V} = 20$
constant $\frac{L}{V}$
 $L = 20V$
 $V = \frac{L}{20} = 0.05 \frac{\text{kmol}}{\text{min}}$
 $y_{N+1} = 0$
pure
water

 $x_N = (1 - 0.91)x_0$
equilibrium
 $y = 28x$

$$a) \quad x_{out} = (1 - 0.91)x_0 = 0.09 (0.012 \text{ mol\%}) = 0.00108 \text{ mol\%}$$

$$x_{out} = 0.000108$$

$$L \cdot x_0 + V \cdot y_{N+1} = L \cdot x_N + V \cdot y_1$$

$$y_1 = \frac{L \cdot x_0 - L \cdot x_N}{V} = \frac{1 \frac{\text{kmol}}{\text{min}} \cdot 0.012 \text{ mol\%} - 1 \frac{\text{kmol}}{\text{min}} \cdot 0.00108 \text{ mol\%}}{0.05 \frac{\text{kmol}}{\text{min}}}$$

$$= 0.2184 \text{ mol\%}$$

$$y_{out} = 0.002184$$

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$$b) \quad N = ? \quad y_1^* = 28x_0$$

$$= 28 \cdot 0.012 \text{ mol\%} = 0.336 \text{ mol\%} = 0.00336$$

$$\frac{L}{mV} = \frac{20}{28}$$

$$y_{N+1}^* = 28 \cdot x_N = 28 \cdot (0.000108) = 3.024 \times 10^{-4}$$

$$N = \frac{\ln(y_{N+1} - y_{N+1}^*) / (y_1 - y_1^*)}{\ln\left(\frac{L}{mV}\right)} = \frac{\ln[0 - 3.024 \times 10^{-4}] / (0.002184 - 0.00336)}{\ln\left(\frac{20}{28}\right)}$$

$$N = \frac{-1.358}{-0.33647} = 4.03636121 \text{ stages}$$

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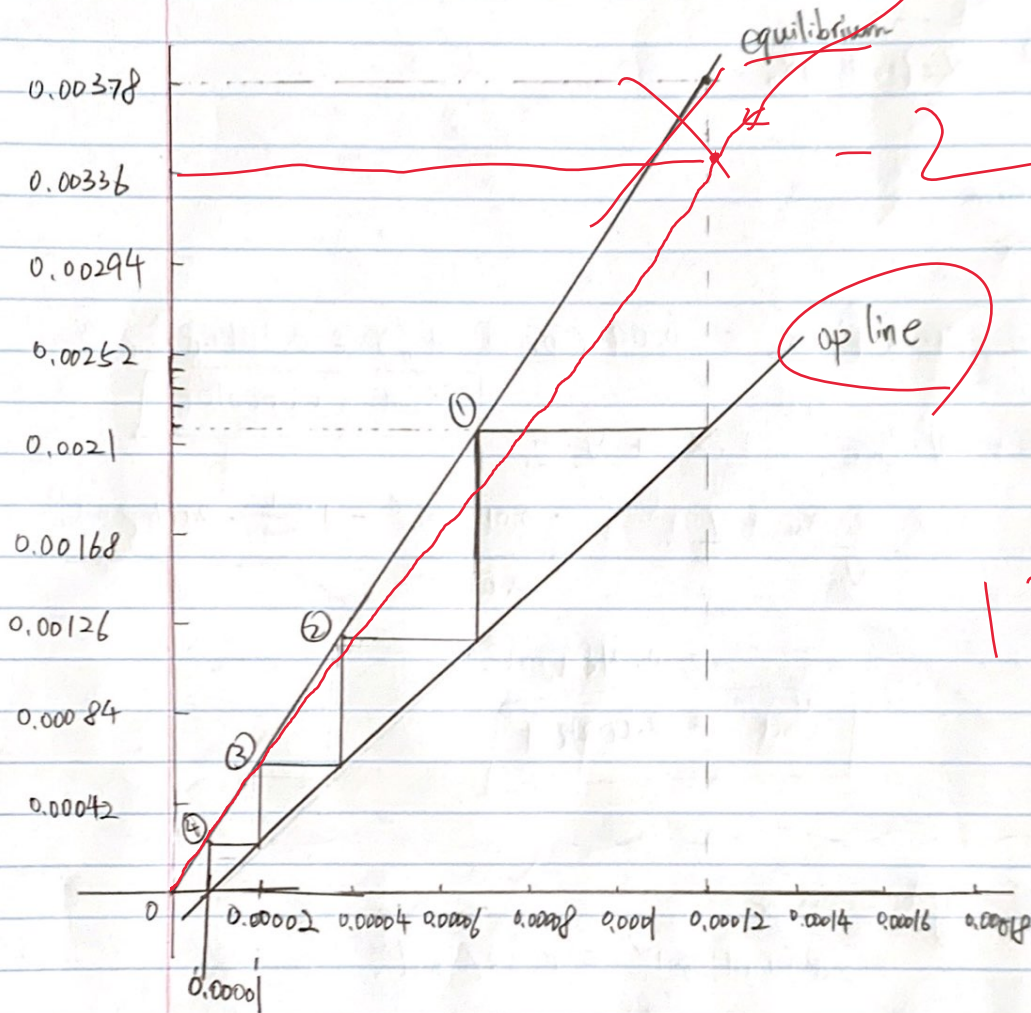
c) equilibrium: $y = 28x$ (0,0) (0.00012, 0.00336) ✓

op line: $y = \frac{L}{V}x + (y_1 - \frac{L}{V}x_0)$

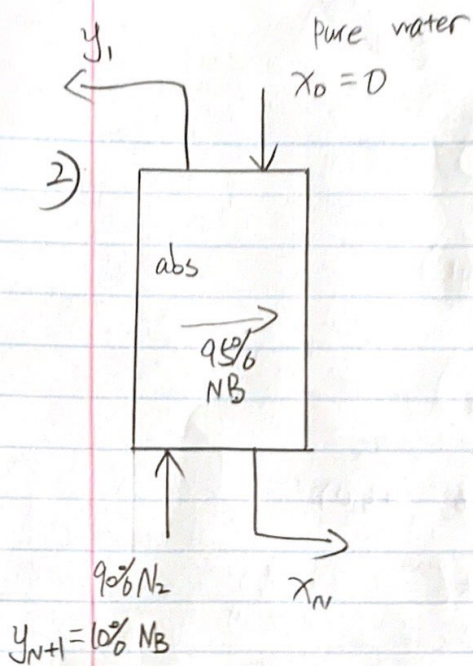
$y = 20x + (0.002184 - 20 \cdot 0.00012)$

$y = 20x - 0.000216$

(0.00012, 0), (0.00012, 0.002184)



④ stages



$$\frac{S}{G} = 45$$

$$y = 28x$$

$$V = 1 \text{ mol/hr}$$

$$a) \quad y_1 = (1 - 0.95) y_{N+1} = 0.05 (0.1) = 0.005$$

$$Y_1 = \frac{y_1}{1 - y_1} = \frac{0.005}{1 - 0.005} = 0.0050251256$$

$$Y_{N+1} = \frac{y_{N+1}}{1 - y_{N+1}} = \frac{0.1}{1 - 0.1} = 0.1111$$

$$X_0 = \frac{x_0}{1 - x_0} = 0$$

$$x_N^* = \frac{Y_{N+1}}{28} = \frac{0.1}{28} = 0.0035714286$$

$$X_N^* = \frac{x_N^*}{1 - x_N^*} = 0.0035842294$$

$$\frac{S}{G}_{\min} = \frac{Y_{N+1} - Y_1}{X_N^* - X_0} = \frac{0.1111 - 0.0050251256}{0.0035842294 - 0} = 29.59767988$$

$$\frac{S}{G} = \frac{Y_{N+1} - Y_1}{X_N - X_0} = 45 \Rightarrow X_N = \frac{Y_{N+1} - Y_1}{45} = \frac{0.1111 - 0.0050251256}{45}$$

$$X_N = 0.0023574417$$

continuous

a) $(X_0, Y_1) = (0, 0.0050251256)$

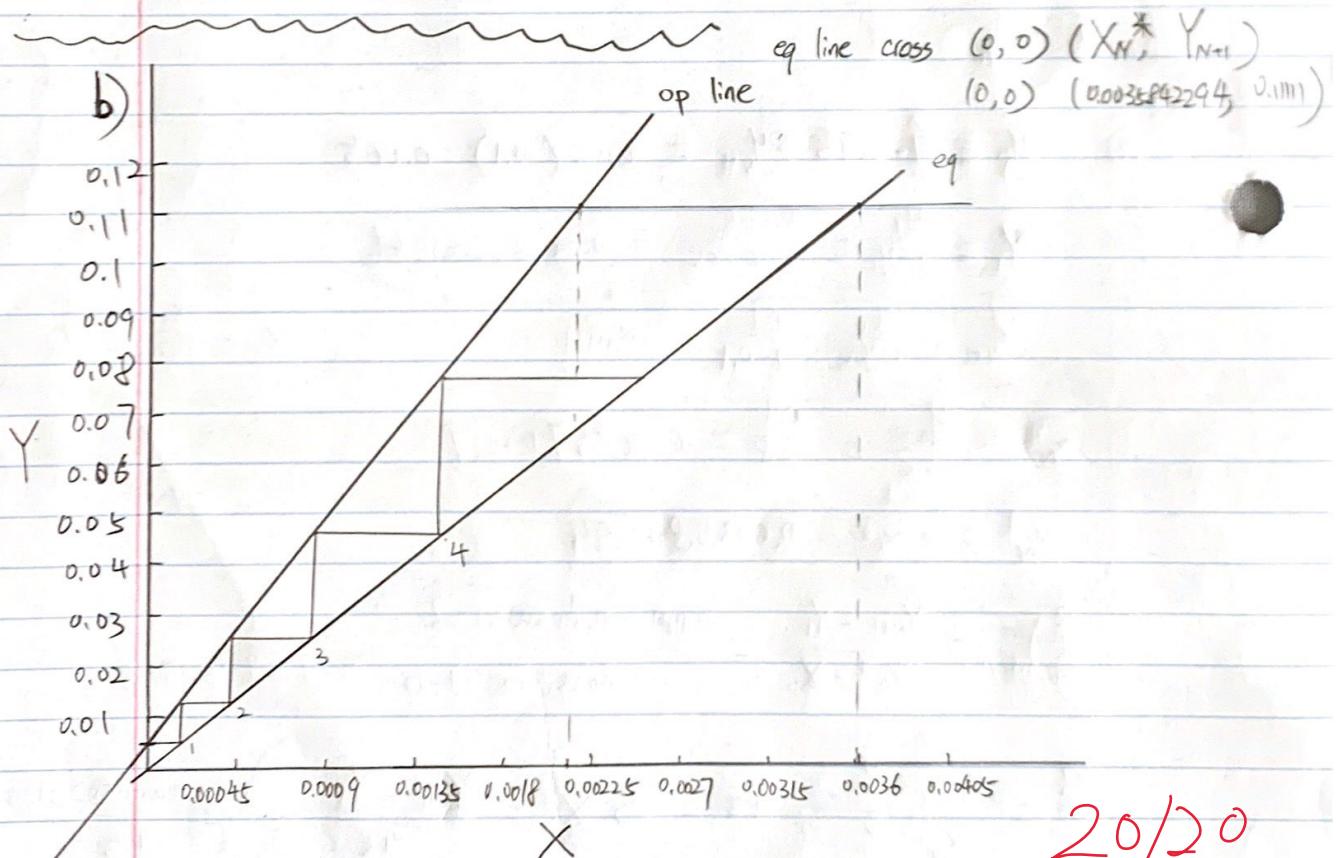
$(X_N, Y_{N+1}) = (0.0023574417, 0.1111)$ ✓

op line: $Y = \frac{S}{G}X + (Y_1 - \frac{S}{G}X_0)$

$Y = 45X + 0.0050251256 - 45(0)$

op line: $Y = 45x + 0.0050251256$ (X_0, Y_1)
 (X_N, Y_{N+1})

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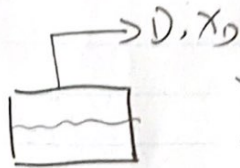
Fraction part: $\frac{7.5}{11} = 0.6818182 + 4 = \# \text{ of stage}$

of stages = 4.6818182

real 5 stages for real

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3)



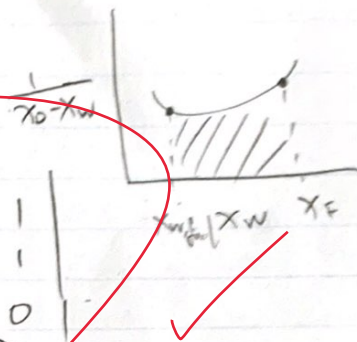
$$X_F = 0.8$$

$$W_{\text{final}} = 2 \text{ kmol}$$

$$X_{W, \text{final}} = 0.4$$

$$F = W_{\text{final}} + D$$

X =	0.2	0.4	0.6	0.8	1
y =	0.7	0.65	0.767	0.925	1
y - x	0.5	0.25	0.167	0.125	0
1 / (y - x)	2	4	5.988	8	



$$\int_{X_{\text{final}}}^{X_F} \frac{1}{y-x} dx = \text{Area under the curve}$$

$$\text{Area} = \frac{0.8 - 0.4}{6} (4 + 4(5.988) + 8) = 2.3968$$

$$W_{\text{final}} = F \cdot e^{-\text{Area}}$$

$$F = \frac{W_{\text{final}}}{e^{-\text{Area}}} = \frac{2 \text{ kmol}}{e^{-2.3968}} = \frac{2 \text{ kmol}}{0.0910087156} = 21.97591722 \text{ kmol}$$

★

$$D = F - W_{\text{final}} = 21.97591722 - 2 \text{ kmol} = 19.97591722 \text{ kmol}$$

Dist
collect

$$D \cdot X_D + W_{\text{final}} \cdot X_{W, \text{final}} = F \cdot X_F$$

$$X_D = \frac{X_F \cdot F - X_{W, \text{final}} \cdot W}{D} = \frac{0.8 \times 21.97591722 - 0.4 \times 2}{19.97591722}$$

$$X_D = 0.840048$$

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